How I validate the train/test data and split it

Question: What is your approach to validating the train/test data and splitting it?

Short answer: I first check that `train-test.csv` and `validation.csv` describe the same world (same cities, coordinates, feature ranges and data problems). Then I split by **time**, not at random: **leave-one-quarter-out cross-validation**. Each fold trains on three quarters of 2025 and tests on the fourth, the way the final model trains on Jan–Oct and predicts Nov–Dec. Everything learned from the data (imputation values, scalers, the outlier formula) is fitted inside each fold on its training rows only. Outliers are removed from training rows only, never from the test quarter.

The IDs in brackets refer to insight_logger.md.

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1. Validating the data before splitting

The model trains on `train-test.csv` (48,000 loads, 2025-01-01 to 2025-10-31) and must predict `validation.csv` (12,000 loads, 2025-11-01 to 2025-12-31). Before choosing a split, I checked whether the two files are consistent.

Check	Result	Decision
Are the cities the same? [THOUGHT1]	64 cities in train-test, 72 in validation. **8 cities appear only in validation** (Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, San Diego) [INS1, INS2]	City names cannot be a reliable feature
Are the routes the same? [THOUGHT2]	536 routes appear only in train-test, **736 only in validation** [INS3]	Drop `pickup` / `delivery` as features [CONC1]; use them only as keys to fill missing values
Are the coordinates consistent?	Every city has the same lat/lon in both files [INS4]	No data-entry problem, but the coordinates add nothing beyond `distance`, so they are dropped [INS5, INS6, CONC1]
Do the same routes behave alike?	On a single route, distance and price still vary from load to load [INS6]	`distance` carries the signal (correlation 0.90 with price [INS10]), not the city name
Missing values	weight 300 / 165 and market_index 374 / 249 (train-test / validation) [INS7]	Same problem in both files, so the same fix applies: fill with the route value, falling back to the global value [THOUGHT5]
Data-quality issues	Weight is capped at 47,500 on **2.5% of rows in both files** [INS8]. Negative weights appear in both [INS9]	Keep the cap, since it occurs in both files and the model can learn it

Check	Result	Decision
		[CONC3]. Apply `abs()` to negative weights [CONC2]
Feature distributions	`market_index` averages **1.08 in train-test and 0.93 in validation**	Market conditions have shifted, so the split must test whether the model copes with a period it hasn't seen (section 2)
Target	`posted_rate` is strongly right-skewed: skew 1.90, and -0.49 after taking the log [INS12]	Train on `log(posted_rate)` [THOUGHT7]
Outliers	677 loads (1.41%) are priced below 0.5× or above 2× the expected price for their distance and equipment [INS13, THOUGHT8]	Remove them from **training** rows only (section 3)

Conclusion: the validation file has the same schema and the same data problems as train-test, but different routes, new cities and lower market prices. A random split would hide these differences, so the split has to be based on time.

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2. The split: leave-one-quarter-out cross-validation

Why not a random split

- **The real task is predicting the future.** The model trains on Jan–Oct and is scored on Nov–Dec. Price levels change month to month (the mean runs from $2,256 in January to $2,497 in June [INS11]). A random split mixes loads from the same days into train and test, so the test score would include no time shift at all.
- **Rows aren't independent.** Loads on the same route in the same week look almost identical. With a random split, near-duplicates land on both sides and the error looks lower than it really is.

What I do instead [THOUGHT6, CONC4, CONC5]

The full date is not a model feature. It is used to **define the folds**: every row gets a `quarter` label, and `LeaveOneGroupOut` holds out one whole quarter at a time.

Fold	Train on	Test on	Test rows
1	2025 Q2, Q3, Q4	2025 Q1	14,291
2	2025 Q1, Q3, Q4	2025 Q2	14,515
3	2025 Q1, Q2, Q4	2025 Q3	14,341
4	2025 Q1, Q2, Q3	2025 Q4 (October only)	4,853

Every test quarter is a period the fold's model has never seen, which is the situation the final model faces on Nov–Dec. Using four folds also shows whether the error is stable across seasons or whether one quarter is

much harder than the others.

The model never sees the year or the month: `date` and `quarter` are dropped before training. From the date, only `day` (day of the month, 1–31) is used, as an ordinal feature.

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3. Keeping the test quarter honest (no leakage)

The full pipeline is rebuilt from scratch in every fold (`sklearn.clone`), so nothing learned on the test quarter leaks into training.

What is learned from data	Where it is fitted
Route median / mean for filling missing weight and market_index	That fold's training rows only
Scalers and the equipment encoder	That fold's training rows only
Outlier formula `ln(price) = start + slope·ln(distance) + equipment shift`	That fold's training rows only
XGBoost model (trained on `log(posted_rate)`)	That fold's training rows, minus their outliers

Outliers are removed from the training rows only. The test quarter keeps every row, including outliers, because real Nov–Dec data will contain them too. For this reason the training CSV *flags* outliers (`is_outlier`) rather than deleting them: deleting them before the split would silently remove the hardest rows from every test quarter.

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4. How each fold is scored

Metric	Meaning
`test_MAE`	Mean absolute error in $ on **all** test rows. This is the main, honest number
`test_MAE_clean`	MAE on test rows that are not outliers. Shows how much of the error comes from outliers
`train_MAE`	MAE on the fold's training rows
`gap`	`test_MAE - train_MAE`. A large gap means overfitting or a quarter that differs from the others

Results (current model, with the `day` feature)

From `freight_rate/model/cv_results.csv`, MAE in $:

Held-out quarter	Rows	Train MAE	Test MAE	Test MAE without outliers	Gap
2025Q1	14,291	41.00	122.74	78.36	81.74
2025Q2	14,515	42.73	112.60	62.93	69.87
2025Q3	14,341	38.47	123.13	71.80	84.66

Held-out quarter	Rows	Train MAE	Test MAE	Test MAE without outliers	Gap
2025Q4	4,853	42.29	139.95	79.89	97.66
Mean		**41.12**	**124.61**	**73.24**	**83.48**

The row-weighted mean test MAE is **121.53**.

Reading the results

- The error is fairly stable across quarters ($113–140), so the model isn't relying on one season.
- About 40% of the test error comes from the 1.4% of rows that are outliers. Without them, the MAE drops from about $125 to about $73.
- Q4 (October only, the fold closest to the Nov–Dec target) has the highest error and the largest gap, which matches the `market_index` shift seen in section 1.

Using the split to judge a feature: `day`

The same cross-validation compares model versions. Adding `day` (day of the month) made the cross-validation error **worse**:

Model	Mean test MAE	Mean test MAE without outliers	Mean train MAE
Without `day`	119.59	68.17	40.39
With `day`	124.61	73.24	41.12

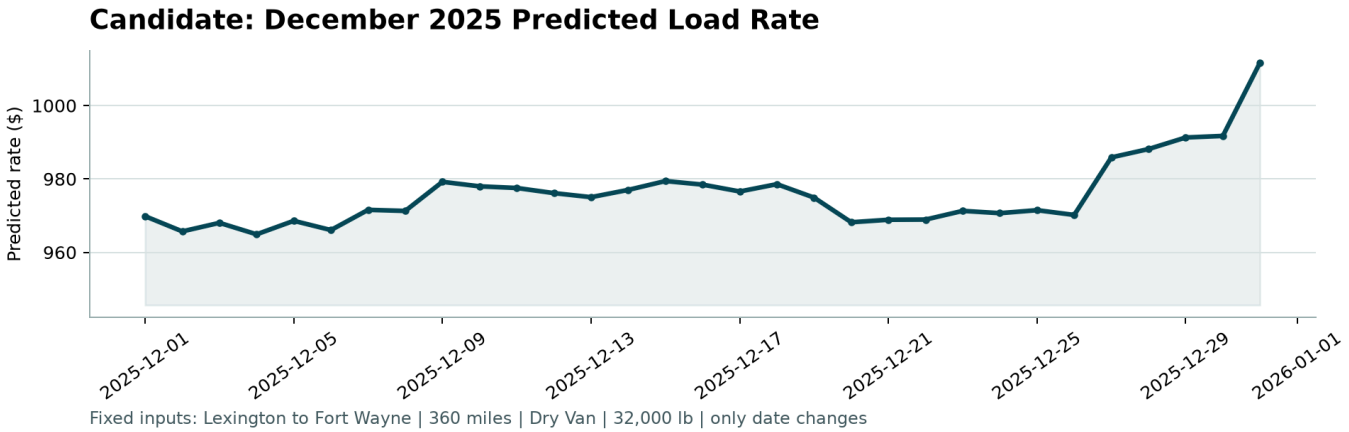
Test error rises in every fold, by $2–8, while train error barely moves. This suggests the model fits day-to-day noise in the training quarters that doesn't repeat in an unseen quarter. Removing `day` would bring back the lower error, but it would also return the December chart in section 5 to a flat line.

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5. The final model and the December chart

After cross-validation, the final model is trained on **all** of Jan–Oct, minus the flagged outliers, and predicts `validation.csv` (Nov–Dec). The data checks from section 1 are applied to prediction data in the same way: `prepare --mode infer` runs the same cleaning, and `score.py` checks the format of the output files.

The December chart predicts one fixed load (Lexington → Fort Wayne, 360 mi, Dry Van, 32,000 lb) for every day of December 2025. Only the date changes, so the shape of the curve comes entirely from the `day` feature:



- The predictions stay between **$965 and $1,012** (mean $976), so day-to-day changes are under 5%.
- The curve is mostly flat around $965–980 for the first four weeks, then rises from December 27 to a peak of $1,012 on December 31.
- The rise at the month's end should be read with caution. Day 31 occurs in only 6 of the 10 training months, so the model has fewer examples for it. As section 4 shows, cross-validation does not confirm that the day pattern generalises.

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6. Limitations

- **Some folds train on later data than they test.** For example, the Q1 fold trains on Q2–Q4. That doesn't happen in production, where the model only ever sees the past. A stricter alternative is forward-chaining (train Q1 → test Q2; train Q1–Q2 → test Q3; …). It matches production more closely but gives fewer, smaller folds. Leave-one-quarter-out was chosen so that every quarter is tested and each model trains on enough data.
- **The quarters have different sizes.** Q4 contains only October (4,853 rows), so its score is noisier. That is why both a plain mean and a row-weighted mean are reported.
- **No fold reproduces the new cities.** Every training quarter already contains all 64 train-test cities, so the cross-validation never tests the 8 cities that appear only in validation. Their rows depend on the global-value fallback for missing data, and their error may be higher than the CV estimate.